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Report Title: Smart Campus Second-hand Trading Platform

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## Amendment History

| Version | Date | Modified By | Reviewed By | Amendment Details |
|---|---|---|---|---|
| 1.0 | 2026/10/18 | Group 11 (SeekDeep) | Group 11 members | Initial version |

# 1. Introduction

## 1.1. Purpose

This document is the User Requirements Specification (URS) for the Smart Campus Second-hand Trading Platform, a web-based marketplace system for students of the Macau University of Science and Technology (MUST). It specifies the functional requirements, non-functional requirements and domain requirements of the system from the users' point of view. It builds on the Task1 Project Proposal and will serve as the input for the High-Level Design (HLD) and Low-Level Design (LLD) in the following tasks.

## 1.2. Notation

This specification and its supporting documents and diagrams use the Unified Modelling Language (UML) notation. Requirement identifiers follow the pattern REQ-x.y for functional requirements, PERF/SEC/AVA/CAP for non-functional requirements, and DOM for domain requirements. Cross-references to Success Criterion N and to the Pair A / Pair B quality commitments refer to the quantified criteria defined in the Task1 Project Proposal [1], Section 4 and Section 3.2.

## 1.3. Scope

The system covers the complete basic workflow of on-campus second-hand trading: user registration and login, product listing management, product search/filter/sort, want-to-buy requests, automatic buyer-seller matching, favorites, transaction status management and in-system notifications.

Out of scope for version 1: online payment, delivery logistics, external identity verification and complex recommendation algorithms. Payment and delivery are agreed offline between the two students; the system only records the transaction status.

## 1.4. Context Diagram

The system is a standalone web application. It interacts with the following external actors and systems:

- Students (buyers and sellers): use the system through a web browser.
- Group 10 (assigned customer group): provides requirements and feedback as the client of Group 11.
- MySQL 8 database: stores all persistent data, hosted on the same server.
- MUST campus e-mail / in-system notification channel: notifies users of matches and status changes.

Figure 1 shows the context diagram of the system: the platform appears as a single node exchanging data with its external actors and systems.

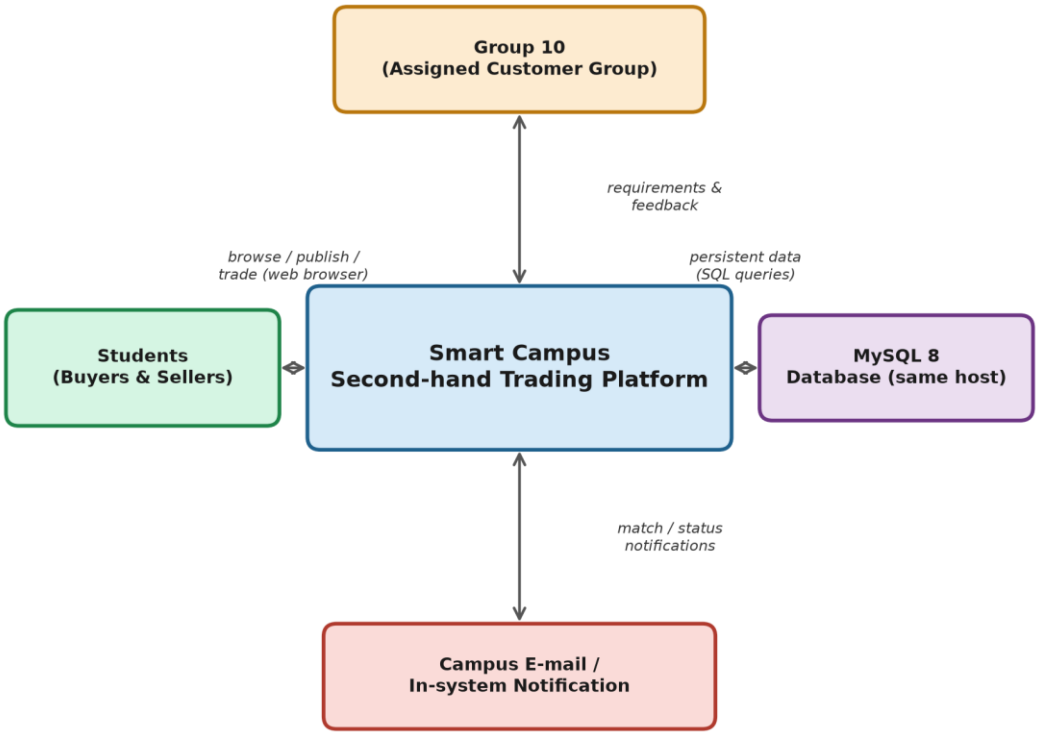



*Figure 1: Context diagram of the Smart Campus Second-hand Trading Platform*

## 1.5. Definitions and Acronyms

| Term | Definition |
|---|---|
| Listing | A product published by a seller, with title, category, description, condition and price. |
| Want-to-buy request | A request published by a buyer describing the desired item, category, budget range and basic requirements. |
| Match | A listing-request pair that satisfies the matching rule (same category, keyword similarity above threshold, asking price within budget). |
| Match score | Weighted score = 0.40*category + 0.35*price overlap + 0.25*condition; only pairs scoring above 0.70 are shown. |
| Reserved / Sold | Transaction status values of a listing after Available. |
| MOP | Macau Pataca, the currency used for all prices in the system. |
| P95 | The 95th percentile response time. |
| UML | Unified Modelling Language. |
| SRS / URS | Software / User Requirements Specification. |

## 1.6. References

[1] Task1 Project Proposal - Smart Campus Second-hand Trading Platform, Group 11, 2026/10/7.

[2] Task2-User Requirements Specifications Template, Software Engineering course, MUST.

[3] Project website / repository: https://github.com/cccc1ccc/smart-campus-secondhand

## 1.7. Overview

Section 2 gives a general description of the system functions and user types. Section 3 and 4 list general constraints, assumptions and dependencies. Section 5 and 6 define the functional requirements in detail (REQ-1.1 to REQ-8.1). Section 7 covers external interfaces. Section 8 covers non-functional requirements. Section 9 states domain requirements. Section 10 describes the requirement elicitation process with the three sources required by the course, and Section 11 lists the contribution of each member to this report.

## 2. General Description

### 2.1. System Functions

The system provides eight high-level functions, inherited from the Task1 proposal and confirmed by requirement elicitation:

- F1 User Registration and Login
- F2 Product Listing Management
- F3 Product Search, Filter and Sort
- F4 Want-to-buy Requests
- F5 Automatic Buyer-Seller Matching (core value of the system)
- F6 Favorites
- F7 Transaction Status Management
- F8 Notifications

### 2.2. User Characteristics

| User Type | Characteristics | Interface Type |
|---|---|---|
| Seller | MUST student who wants to sell used items (textbooks, electronics, household items). Occasional user, no training. Uses a PC or mobile browser on campus network. | Self-service web UI |
| Buyer | MUST student looking for cheaper second-hand items. May post want-to-buy requests. Occasional user, no training. | Self-service web UI |
| Both roles | The same student account can act as seller and buyer at the same time; no separate registration is required. | Self-service web UI |
| Team / admin (developer) | Group 11 members who maintain the system, seed test data and monitor logs. | Admin console / database access |

All users are expected to read English or Chinese; version 1 provides an English user interface. No data-entry training is needed; every form contains at most five fields.

## 3. General Constraints

### 3.1. Software Constraints

- The system must be implemented with Java 17 and Spring Boot, using Maven as the build tool and JUnit 5 for unit testing (course-approved stack fixed in Task1).
- Persistent data must be stored in MySQL 8.
- The system must be a web application usable from current Chrome and Edge browsers.

- Source code must be hosted on the public GitHub repository of the project.
- Only free or open-source libraries may be used; no licence cost is allowed.

### 3.2. Hardware Constraints

- The system must run on a single ordinary PC or a free-tier cloud host; no dedicated server hardware is available.
- Users access the system from ordinary laptops or mobile phones; no special client hardware is required.

## 4. Assumptions and Dependencies

- A1: Users are MUST students who can access the campus network and use a web browser.
- A2: Payment and delivery are completed offline between the two students; the system only records the transaction status.
- A3: The matching rule in F5 is rule-based and transparent; no external AI service is used.
- A4: The assigned customer group (Group 10) is available for requirement confirmation meetings.
- D1: The system depends on MySQL 8 and a Java 17 runtime on the host machine.
- D2: The demonstration depends on a seeded database of at least 5,000 listings for the performance benchmark.

## 5. Functional Requirements Master List

The table below lists all functional requirements in summary form. Each requirement is defined in detail in Section 6.

| Req. ID | Requirement Name | Requirement Description |
|---|---|---|
| REQ-1.1 | User registration | A visitor can create an account with name, student number, e-mail and password. |
| REQ-1.2 | User login and profile | A registered user can log in and maintain basic profile information. |
| REQ-2.1 | Publish listing | A seller can publish a product with title, category, description, condition and price. |
| REQ-2.2 | Manage listing | A seller can edit an unpublished listing or mark it as sold. |
| REQ-3.1 | Search, filter and sort | A buyer can search by keyword and filter/sort by category, price and condition. |
| REQ-4.1 | Publish want-to-buy request | A buyer can publish a request with the desired item, category, budget range and requirements. |
| REQ-5.1 | Automatic matching | The system compares active listings with want-to-buy requests and reports matches above the score threshold. |
| REQ-6.1 | Favorites | A user can save and remove products in a favorites list. |
| REQ-7.1 | Transaction status | A user can track and update the status of an item: Available, Reserved, Sold. |
| REQ-8.1 | Notifications | A user receives in-system notifications when a match is found or a relevant status changes. |

# 6. Functional Requirements (Detailed)

## 6.1 Functional Requirement REQ-1.1 User Registration

**Description:** A visitor creates an account. The account identifies the user as a MUST student and is the precondition for publishing listings, requests and favorites.

**System Input:** Name, student number, e-mail address, password (entered twice).

**Display:** A registration form with clearly labelled fields, inline validation messages and a submit button. At most five input fields.

**System Processing:** The system validates that all fields are non-empty, the e-mail format is valid, the student number is unique, and the two password entries are identical. The password is stored only as a salted hash. On success the account is created and the user is logged in.

**System Output:** A welcome page and a confirmation message; the user is redirected to the home page.

**Other:** None.

**Constraints:** Registration must be possible within 2 minutes for a new user (part of Success Criterion 1).

**Data Handling:** All strings are stored in UTF-8 (Unicode). E-mail and student number are validated before the account is persisted.

**Error Handling:** Duplicate student number or e-mail: reject with a specific message. Weak password (shorter than 8 characters): reject and ask for a stronger password.

## 6.2 Functional Requirement REQ-1.2 User Login, Profile and Account Deletion

**Description: A registered user logs in, maintains basic profile information and may delete the account**.

**System Input:** E-mail / student number and password for login; editable profile fields (name, contact, password change with old password).

**Display:** Login form; profile page showing current data with edit capability.

**System Processing:** The system verifies the credential against the stored salted hash. Failed attempts are counted; after 5 consecutive failures the account is locked for 10 minutes. Profile updates are validated as in REQ-1.1. On confirmed account deletion, the system removes the user profile, listings, want-to-buy requests and favorites, and also removes the favorites and match records other users created for the deleted account listings.

**System Output: On success the home page with the user session; after account deletion the user is logged out**; on failure a clear error message.

**Other:** Session expires after 2 hours of inactivity.

**Constraints: Account deletion requires explicit confirmation**.

**Data Handling: Password change requires the old password; account deletion is performed in one transaction so a failure never leaves partial data. The new password** must differ from the old one.

**Error Handling:** Wrong credentials: generic 'invalid credential' message (no hint which field failed). Locked account: message with remaining lock time.

### 6.3 Functional Requirement REQ-2.1 Publish Listing

**Description:** A seller publishes a second-hand product so that buyers can find it.

**System Input:** Title, category (from a fixed list), description, condition (new / like-new / good / fair), asking price in MOP.

**Display:** A publishing form with at most five form fields and completion in at most three clicks (Pair A quality commitment).

**System Processing:** The system validates the price as a positive number, stores the listing with status Available and the current timestamp, and immediately triggers the matching process of REQ-5.1 against all active want-to-buy requests.

**System Output:** The published listing page and a success confirmation; the listing appears in search results.

**Other:** Photos are optional in version 1 (one photo per listing, JPEG/PNG, at most 5 MB).

**Constraints:** Only logged-in users can publish. A user can have at most 50 active listings.

**Data Handling:** Title limited to 100 characters, description to 2,000 characters, price with two decimal places in MOP.

**Error Handling:** Missing mandatory field: highlight the field and refuse submission. Oversized photo: reject with size hint.

### 6.4 Functional Requirement REQ-2.2 Manage and Delete Listing

**Description: A seller edits or deletes an existing listing, or marks it as sold**.

**System Input:** Listing ID (selected from the seller's own listing list) plus the fields to change.

**Display:** A 'My listings' page showing the seller's listings with status, and an edit page identical in layout to the publish form.

**System Processing:** The system checks that the current user owns the listing. Edits are allowed only while the listing is Available. Marking as Sold sets the final status and removes the listing from matching and default search results. Deleting a listing removes it together with all favorites that reference it and all of its match records; related notifications are kept but marked as referring to a removed listing.

**System Output:** Updated listing page with a success confirmation.

**Other:** None.

**Constraints:** A listing marked as Sold cannot be edited or restored in version 1.

**Data Handling: Every update is timestamped; deletion is performed in one transaction together with the cleanup of favorites and match records**; the edit history is not kept in version 1.

**Error Handling: Editing or deleting a listing owned by another user** or a sold listing: refuse with an explanatory message.

### 6.5 Functional Requirement REQ-3.1 Search, Filter and Sort

**Description:** A buyer searches products by keyword and filters or sorts the results by category, price range and condition.

**System Input:** Keyword (optional), category, minimum and maximum price, condition, and sort order (newest, price ascending, price descending).

**Display:** A result list with thumbnail, title, price, condition and status, grouped by page (20 items per page). Filters are shown as applied chips that can be removed individually.

**System Processing:** The system returns only listings with status Available that satisfy every selected condition. Keyword matching covers title and description. P95 response time must be under 2 seconds on a seeded database of 5,000 listings (Pair A quality commitment and Success Criterion 5).

**System Output:** The filtered and sorted result list; an empty-result message with suggestions when nothing matches.

**Other:** None.

**Constraints:** Search results never include listings of status Sold.

**Data Handling:** Numeric filters are validated; min price greater than max price is rejected.

**Error Handling:** Invalid filter combination: show a message and keep the previous results.

### 6.6 Functional Requirement REQ-4.1 Publish Want-to-buy Request

**Description:** A buyer publishes a want-to-buy request describing the desired item so that the matching function can find suitable listings.

**System Input:** Desired item keywords, category, budget range (min-max in MOP), and basic requirements (e.g. acceptable condition).

**Display:** A request form with the same style as the listing form, plus a 'My requests' list page.

**System Processing:** The system validates the budget range, stores the request as active, and immediately triggers the matching process of REQ-5.1 against all active listings. Requests can be edited or closed by the owner.

**System Output:** The published request page and a success confirmation.

**Other:** None.

**Constraints:** A user can have at most 20 active requests.

**Data Handling:** Budget range must be a valid positive numeric interval.

**Error Handling:** Invalid budget interval: reject with a message; other errors follow the common error handling of Section 6.13.

### 6.7 Functional Requirement REQ-5.1 Automatic Buyer-Seller Matching

**Description:** The core function of the system. The system compares active listings with active want-to-buy requests and identifies potential matches automatically, without any manual search by the user.

**System Input:** All active listings and all active want-to-buy requests (no user input at match time).

**Display:** A 'Matches' page per user showing each matched pair (listing + request) with its score, sorted by score descending.

**System Processing:** A pair is a candidate when the categories are the same, the keyword similarity is above a set threshold, and the asking price falls inside the buyer's budget range. Candidates are ranked by the weighted score: score = 0.40*category + 0.35*price overlap + 0.25*condition. Only pairs scoring above 0.70 are shown to the user. Matching is re-run whenever a listing or request is created, edited or closed. Quality targets (Success Criteria 3): recall at least 85% and top-3 precision at least 70% on a hand-labelled set of 50 request-listing pairs.

**System Output:** The ranked match list for the affected users, and a notification per REQ-8.1 for each new qualifying match.

**Other: The scoring rule and its weights are documented and shown to users for transparency. One listing may match several want-to-buy requests at the same time: every qualifying listing-request pair is an independent match shown separately in score order, so the seller can review all interested buyers and choose whom to trade with offline. Matches are not exclusive; when a listing becomes Reserved or Sold, its remaining open matches are closed and the affected buyers are notified per REQ-8.1**.

**Constraints:** The rule and thresholds are fixed for version 1; no AI or external recommendation service is used.

**Data Handling:** Similarity is computed on normalised (lower-cased, trimmed) text; all amounts are in MOP. When a listing or request is deleted, its match records are removed together with it (see REQ-2.2 and REQ-1.2).

**Error Handling:** If the matching process fails (e.g. database error), the listing/request is still saved and matching is retried by a scheduled re-run; the failure is logged.

## 6.8 Functional Requirement REQ-6.1 Favorites

**Description:** A user saves products that he or she may want to purchase later.

**System Input:** Listing ID (via a 'save' button on any listing page).

**Display:** A 'Favorites' page listing all saved products with price and status, most recently saved first.

**System Processing:** The system adds or removes the listing in the user's favorites list. Duplicate saves of the same listing are ignored. When a favorited listing changes status or price, a notification is created per REQ-8.1.

**System Output:** Updated favorites page and a confirmation of the add/remove action.

**Other:** None.

**Constraints:** A user can save at most 200 favorites.

**Data Handling:** Favorites reference the listing ID only; no data is duplicated.

**Error Handling:** Saving a sold or deleted listing: refuse with a message.

## 6.9 Functional Requirement REQ-7.1 Transaction Status Management

**Description:** Users track and update the basic status of an item through the trade: Available, Reserved, Sold.

**System Input:** Listing ID and the target status (selected by the seller from the allowed transitions).

**Display:** The status is displayed as a coloured badge on the listing page, in search results, in favorites and in 'My listings'.

**System Processing:** Allowed transitions: Available to Reserved, Reserved to Sold, Reserved back to Available. The seller performs the transition; the system records the timestamp and notifies the users involved (matched buyer, users who favorited the listing) per REQ-8.1.

**System Output:** The updated status shown everywhere the listing appears.

**Other:** None.

**Constraints:** Invalid transitions (e.g. Sold to Available) are not allowed in version 1.

**Data Handling:** Status history (timestamped transitions) is stored for the lifetime of the listing.

**Error Handling:** Transition by a user who is not the seller, or an invalid transition: refuse with an explanatory message.

## 6.10 Functional Requirement REQ-8.1 Notifications

**Description:** Users receive in-system notifications when a potential match is found or when a relevant item status changes.

**System Input:** System events: new qualifying match (REQ-5.1), status change (REQ-7.1), price change of a favorited listing (REQ-6.1).

**Display:** A notification bell with an unread counter, and a notification list page with title, short text, link to the related listing/request, timestamp and read/unread state.

**System Processing:** For each event the system creates one notification per affected user. A qualifying match must produce a notification to both users within 60 seconds of the match being found (Pair B quality commitment and Success Criterion 4). Notifications can be marked as read individually or all at once.

**System Output:** The notification entries in the user's notification list; the unread counter is updated in real time on the next page load.

**Other:** E-mail notification is optional in version 1 and may be disabled.

**Constraints:** Notifications are kept for 90 days, then deleted automatically.

**Data Handling:** Notification text is a short template in English with the listing title and amount inserted.

**Error Handling:** If notification creation fails, the event is logged and retried once; the business operation itself is not rolled back.

6.11-6.12: reserved. 6.13 Common error handling: all user-facing errors return a clear message near the triggering field; unexpected system errors show a generic message and are written to the server log with timestamp and user ID.

# 7. External Interface Requirement

## 7.1. Data Interfaces

The system stores all data in a MySQL 8 database on the same host. There is no external data feed in version 1. Test seed data (at least 5,000 listings) is loaded from SQL scripts kept in the project repository.

## 7.2. User Interfaces

A responsive web user interface usable from current Chrome and Edge browsers on laptops and mobile phones. Pages: home/search, listing detail, publish/edit forms, want-to-buy requests, matches, favorites, my listings, notifications, login/registration and profile. All forms use inline validation and English labels.

## 7.3. Other Interfaces

No hardware or communication interfaces beyond HTTPS between the browser and the server. Optional e-mail notification, if enabled, uses the host's SMTP service; the system works fully without it.

# 8. Non-Functional Requirements

## 8.1. System Performance

| Req. Id | Description | Response Time |
|---|---|---|
| PERF-1.1 | Time for regular operations such as moving between fields or pages of the same screen. | 1 second |
| PERF-1.2 | Search, filter and sort on a seeded database of 5,000 listings (P95). | 2 seconds |
| PERF-1.3 | Publishing a listing or a want-to-buy request including validation and persistence. | 3 seconds |
| PERF-1.4 | Automatic matching re-run after a create/edit/close event, per affected user set. | 10 seconds |
| PERF-1.5 | Notification delivered after a qualifying match is found. | 60 seconds |

## 8.2. Information Security

| Req. Id | Description |
|---|---|
| SEC-1.1 | Passwords must never be stored or transmitted in plain text; only salted hashes are stored, and all traffic uses HTTPS. |
| SEC-1.2 | Personal data (name, student number, e-mail) is visible only to the owner; other users see only the display name on a listing. |

| | |
|---|---|
| SEC-1.3 | A user can modify or delete only his or her own listings, requests and favorites; every modifying request is authorised server-side. |
| SEC-1.4 | After 5 consecutive failed logins the account is locked for 10 minutes. |
| SEC-1.5 | All user input is validated server-side; SQL access uses parameterised queries to prevent injection. |

### 8.3. Availability

| Req. Id | Description |
|---|---|
| AVA-1.1 | The system is available during normal campus hours (08:00-24:00) on all teaching days. |
| AVA-1.2 | Planned maintenance is announced at least 24 hours in advance and does not exceed 2 hours per week. |
| AVA-1.3 | Unplanned downtime should not exceed 4 hours per month during the demonstration period. |
| AVA-1.4 | After a crash, the system restarts with no loss of committed data (database transactions). |

### 8.4. Capacity

| Req. Id | Description |
|---|---|
| CAP-1.1 | Year ONE data volume: up to 5,000 listings, 1,000 want-to-buy requests and 500 registered users. |
| CAP-2.1 | Year ONE concurrency: at least 50 concurrent users searching without violating PERF-1.2. |
| CAP-3.1 | Favorites: up to 200 per user; active listings per user: 50; active requests per user: 20; notifications retained 90 days. |

Note on capacity planning: Year TWO and Year THREE volumes are out of scope for version 1. The platform is delivered as a course prototype sized for the Year ONE data and concurrency figures in CAP-1.1 and CAP-2.1; growth beyond these figures will be re-planned in a later version if the service is adopted by more students.

## 9. Domain Requirements

| Req. Id | Description |
|---|---|
| DOM-1.1 | All prices and budgets are expressed in MOP (Macau Pataca) with two decimal places; no currency conversion is provided. |
| DOM-1.2 | The system serves only the MUST campus community; the item category list reflects typical campus goods (textbooks, electronics, household items, sports gear, others). |
| DOM-1.3 | Transaction status vocabulary is fixed to Available / Reserved / Sold; payment and delivery are offline and out of scope. |
| DOM-1.4 | Personal data is handled according to the Macao SAR Personal Data Protection Act (Law No. 8/2005): data is used only for the trading function and deleted with the account. |
| DOM-1.5 | The matching rule must remain transparent and explainable to users (weights and threshold are published). |

## 10. Requirement Elicitation Process

Requirements were elicited from the three sources required by the course:

### 10.1. Method 1 - Assigned Customer Group (compulsory)

Group 10 acts as the customer of Group 11. An elicitation meeting was held in which Group 10 reviewed the F1-F8 feature list from the Task1 proposal and asked questions from the user's point of view. Main outcomes: (a) the matching rule must be explainable, not a black box (adopted as DOM-1.5 and the transparency note in REQ-5.1); (b) publishing a listing must be fast, so the form is limited to five fields and three clicks (REQ-2.1); (c) notifications must arrive quickly, set as the 60-second target (REQ-8.1, PERF-1.5). Group 11 also acts as the customer of Group 12.

### 10.2. Method 2 - Real Potential Users (recommended)

Three to five MUST students who actually buy or sell second-hand items were interviewed. Main findings: scattered chat-group information is the biggest pain point (supports F3 and F5); students want price filtering and MOP display (REQ-3.1, DOM-1.1); sellers want a simple Reserved status to avoid repeated answers (REQ-7.1). Interview notes are kept in the project repository.

### 10.3. Method 3 - Team Members as Surrogate Users (optional)

All four team members are MUST students and second-hand traders themselves. From this perspective the team added the favorites function (F6) and the limits on active listings and requests (CAP-3.1) to keep the system clean.

## 11. Contributions to This Report

This report was produced jointly by the four members of Group 11. The table below states the specific contribution of each member to this Task2 - User Requirements Specifications report. Each member contributed approximately 25% of the total work.

| Member | Student No. | Contribution to This Report | Share |
|---|---|---|---|
| Wu Yutong | 1240016186 | Led the team; organised the elicitation meeting with Group 10; wrote Sections 1, 3 and 4 (Introduction, Constraints, Assumptions); compiled and finalised the complete report. | 25% |
| Wang Yike | 1240012305 | Conducted the user interviews (Method 2); wrote Section 2 (General Description) and the detailed requirements REQ-1.1 to REQ-3.1; maintained the master list in Section 5. | 25% |
| Chen Moxiong | 1240008865 | Specified the matching requirement REQ-5.1 with its scoring rule and quality targets; wrote REQ-4.1 and Section 8 (Non-Functional Requirements). | 25% |
| Gao Shengzhe | 1240008949 | Wrote REQ-6.1 to REQ-8.1, Sections 7, 9 and 10 (Interfaces, Domain Requirements, Elicitation Process); applied the template formatting and | 25% |

| | | | |
|---|---|---|---|
| | | proofread the final document. | |